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# Calculate additional statistics
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1. skewness = apple['Returns'].skew()
2. kurtosis = apple['Returns'].kurt()
3. serial_corr = apple['Returns'].autocorr()
4.
5. print(f"Skewness: {skewness:.4f}")
6. print(f"Kurtosis: {kurtosis:.4f}")
7. print(f"Serial Correlation: {serial_corr:.4f}")
8.
```

For Apple during this period, we might see:

Skewness: -0.5289

Kurtosis: 4.2731

Serial Correlation: -0.0354

These numbers would tell us that Apple's returns have:

- Negative skew (more extreme negative moves than positive)
- Excess kurtosis (more extreme moves than a normal distribution)
- Very slight negative serial correlation (a slightly higher chance of reversal than continuation)

These insights would be crucial for strategy development—suggesting that risk management for extreme downside events would be particularly important when trading this stock.

Data Visualization: Making Numbers Come Alive

While numbers are powerful, humans are visual creatures. Visualizing data can reveal patterns that remain hidden in tables of statistics. Let's explore the key visualization techniques that traders use to analyze financial data.

Line Charts: Tracking Price Evolution

The line chart is the most basic and essential visualization in trading. It shows the evolution of price (or any other variable) over time, revealing trends, cycles, and key levels.